Question	Answer	Marks	AO	Guidance
4	AE $= 4\cos x°$	**M1***	**3.1a**	Uses basic trig to find an expression for either AE or EC
	EC $= 7\sin x°$			
	E is midpoint so $4\cos x° = 7\sin x°$	**M1**	**3.1a**	Equates their expressions
	So $\dfrac{\sin x°}{\cos x°} = \tan x° = \dfrac{4}{7}$	**M1(dep)**	**1.2**	Uses a correct trig identity leading to a value for $\tan x$ or equivalent
	So $x = 29.7$	**A1**	**1.1**	cao
	Alternative method			
	Triangles BAE and CDE are similar with scale factor 1.75	**M1***		Identifying similar triangles and the scale factor
	Let AE = y cm = CE			
	DE = $1.75y$	**M1**		Uses scale factor to find expression for DE or BE as $\frac{4}{7}y$
	$\tan x° = \dfrac{CE}{DE} = \dfrac{y}{1.75y} = \dfrac{4}{7}$	**M1(dep)**		Uses basic trig ratio Also allow use of Pythagoras and another trig ratio instead.
	So $x = 29.7$	**A1**		cao
		[4]		